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STEPCONE-2025

STANDARD OPERATING PROCEDURE

EVENT NAME: Circuit Debugging

DEPARTMENT NAME: Electronics and Communication Engineering (ECE)

EVENT TYPE: Technical Event

EVENT DESCRIPTION:

This contest invites electronics enthusiasts to test their skills in identifying and fixing faults in various circuits. Participants will diagnose intentional errors in a series of circuits, with each round increasing in complexity and requiring quick thinking, precision, and deep electronics knowledge. The goal is to debug each circuit within the time limit, showcasing expertise and gaining hands-on troubleshooting experience in a high-energy, competitive setting.

Round 1:

1. Participants need to answer for the basic questions related to electronics which were asked by the organizing team.
2. Based on the cutoff points teams will be qualified for next round.

Round 2:

1. For the qualifiers, few circuit diagrams with faults will be given, then they need to find errors.
2. The team who solves more faults within a given time limit will be received components which can be helpful for them to crack the next round.

Round 3:

1. Qualified teams need to complete the given circuit with the help of given components in 2nd round and they need to show the final output.
2. Whoever solves first they will be declared as winners.

EVENT GUIDELINES:

1. Eligibility:

- Open to all electronics students, enthusiasts, and professionals interested in circuit debugging.

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- Participants may enter individually or as teams.

2. Tools and Equipment

- ***Provided Tools:*** Multimeters, oscilloscopes, breadboards, power supplies, and essential components will be available at each workstation.
- ***Personal Tools:*** Participants may not use external tools or components unless specified by the organizers.

3. Rules and Conduct

- ***Time Limits:*** Each round has a fixed time limit. Participants must submit their solutions before time expires.
- ***Safety Protocols:*** Participants must follow safety protocols when handling tools and components.
- ***Penalty:*** Incorrect fixes or failure to explain solutions may lead to point deductions.

Team Size/No of Participants: 2 per team

PLAN OF ACTION: Day-shift

Duration of event:

4 hours – Day 2 - 10:00 AM to 3:00 PM

Judging Criteria: Based on accuracy, speed and explanation of fault finding.